

Action

A Schema.org Type

Thing > **Action**

[\[more...\]](#)

An action performed by a direct agent and indirect participants upon a direct object. Optionally happens at a location with the help of an inanimate instrument. The execution of the action may produce a result. Specific action sub-type documentation specifies the exact expectation of each argument/role.

See also [blog post](#) and [Actions overview document](#).

Property	Expected Type	Description
Properties from Action		
actionProcess	HowTo	Description of the process by which the action was performed.
actionStatus	ActionStatusType	Indicates the current disposition of the Action.
agent	Organization or Person	The direct performer or driver of the action (animate or inanimate). E.g. <i>John</i> wrote a book.
endTime	DateTime or Time	The endTime of something. For a reserved event or service (e.g. FoodEstablishmentReservation), the time that it is expected to end. For actions

Property	Expected Type	Description
		that span a period of time, when the action was performed. E.g. John wrote a book from January to <i>December</i>. For media, including audio and video, it's the time offset of the end of a clip within a larger file. Note that Event uses startDate/endDate instead of startTime/endTime, even when describing dates with times. This situation may be clarified in future revisions.
error	Thing	For failed actions, more information on the cause of the failure.
instrument	Thing	The object that helped the agent perform the action. E.g. John wrote a book with <i>a pen</i> .
location	Place or PostalAddress or Text or VirtualLocation	The location of, for example, where an event is happening, where an organization is located, or where an action takes place.
object	Thing	The object upon which the action is carried out, whose state is kept intact or changed. Also known as the semantic roles patient, affected or undergoer (which change their state) or theme (which doesn't). E.g. John read <i>a book</i> .
participant	Organization or Person	Other co-agents that participated in the action indirectly. E.g. John wrote a book with <i>Steve</i> .
provider	Organization or Person	The service provider, service operator, or service performer; the goods producer. Another party (a seller) may offer those services or goods on behalf of the provider. A provider may also serve as the seller. Supersedes carrier .
result	Thing	The result produced in the action. E.g. John wrote <i>a book</i> .

Property	Expected Type	Description
startTime	DateTime or Time	The startTime of something. For a reserved event or service (e.g. FoodEstablishmentReservation), the time that it is expected to start. For actions that span a period of time, when the action was performed. E.g. John wrote a book from <i>January</i> to December. For media, including audio and video, it's the time offset of the start of a clip within a larger file. Note that Event uses startDate/endDate instead of startTime/endTime, even when describing dates with times. This situation may be clarified in future revisions.
target	EntryPoint or URL	Indicates a target EntryPoint, or url, for an Action.
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This is a relationship between something and a class that the thing is in. Typically the value is a URI-identified RDF class, and in this case corresponds to the use of rdf:type in RDF. Text values can be used sparingly, for cases where useful information can be added without their being an appropriate schema to reference. In the case of text values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other properties (in

Property	Expected Type	Description
		particular, name) may be necessary for the description to be useful for disambiguation.
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of Thing , such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.
sameAs	URL	URL of a reference Web page that unambiguously indicates the item's identity. E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website.
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Instances of **Action** may appear as a value for the following properties

Property	On Types	Description
interactionType	InteractionCounter	The Action representing the type of interaction. For up votes, +1s, etc. use LikeAction . For down votes use DislikeAction . Otherwise, use the most specific Action.
potentialAction	Thing	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.

More specific Types

- **AchieveAction**
- **AssessAction**
- **ConsumeAction**
- **ControlAction**
- **CreateAction**
- **FindAction**
- **InteractAction**
- **MoveAction**
- **OrganizeAction**
- **PlayAction**
- **SearchAction**
- **SeekToAction**
- **SolveMathAction**
- **TradeAction**
- **TransferAction**

- **UpdateAction**

Acknowledgements

Hydra project

The schema.org Actions mechanism benefited from extensive discussions across the Web standards community around W3C, in particular from the **Hydra project**'s community group.

Examples

📄 Example 1

No Markup

JSON-LD

Structure

Example encoded as JSON-LD in a HTML script tag.

```
<!-- John listened to Pink with Steve at Anna's apartment on his iPod. -->
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "ListenAction",
  "agent": {
    "@type": "Person",
    "name": "John"
  },
  "object": {
    "@type": "MusicGroup",
```

```
    "name": "Pink!",
  },
  "participant": {
    "@type": "Person",
    "name": "Steve"
  },
  "location": {
    "@type": "Residence",
    "name": "Ann's apartment"
  },
  "instrument": {
    "@type": "Product",
    "name": "iPod"
  }
}
</script>
```

📄 Example 2

No Markup

JSON-LD

Structure

Example encoded as JSON-LD in a HTML script tag.

```
<!-- John created an exercise plan with Steve. -->
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "CreateAction",
```

```
"agent": {
  "@type": "Person",
  "name": "John"
},
"result": {
  "@type": "ExercisePlan",
  "name": "John's weight loss plan"
},
"participant": {
  "@type": "Person",
  "name": "Steve"
}
}
</script>
```